

CSCE 753 COMPUTER VISION · MODULE 03 · LECTURE NOTES

Features and Correspondence

The problem everything geometric depends on.

MODULE OUTCOMES

- Explain what makes a good feature point.
- Explain scale and rotation invariance and how it is achieved.
- Match descriptors and filter the matches.
- Apply RANSAC and choose its parameters.
- Explain why learned matchers improved on hand-designed ones.

1 What makes a good feature

A GOOD FEATURE IS ONE YOU CAN LOCALISE IN TWO DIMENSIONS

A flat patch matches everywhere in a uniform region, giving no information about position at all. Nothing can be done with it.

An edge patch matches anywhere along the edge. One degree of freedom is determined and one is not. **This is the aperture problem**, and it returns in Module 07 as the reason optical flow is under-determined at every pixel individually — the same geometry, in a different application.

A corner matches in exactly one place. Intensity varies in both directions, so both image coordinates are pinned down, which is what correspondence requires.

Harris formalises this as the eigenvalues of the local gradient covariance matrix: two large eigenvalues is a corner, one large is an edge, neither is a flat region — and the corner response is a cheap function of those eigenvalues that avoids computing them (§1's code).

Consider shifting a window by (dx, dy) . To first order the intensity change is

$$E(dx, dy) = [dx \ dy] \ M \ [dx \ dy]^T$$

where M is the GRADIENT COVARIANCE (the structure tensor) over the window:

$$M = \text{sum_w} \begin{bmatrix} I_x * I_x & I_x * I_y \\ I_x * I_y & I_y * I_y \end{bmatrix}$$

M 's eigenvalues say how much the patch changes when moved along each of two principal directions:

both small	-> flat region, moves freely:	USELESS
one large	-> edge, one free direction:	PARTIAL
both large	-> corner, pinned in 2D:	GOOD

Harris avoids an eigendecomposition:

$$R = \det(M) - k * \text{trace}(M)^2, \quad k \text{ about } 0.04 \text{ to } 0.06$$

which is large and positive only when both eigenvalues are large. Threshold R , then apply non-maximum suppression.

NOTE THE STRUCTURE: a 2x2 covariance whose eigenvalues say how well-determined a local estimate is. That is the condition number of Module 01 section 2, computed locally.

The same matrix appears in Module 07's Lucas–Kanade optical flow, where its conditioning decides whether flow is recoverable at a given pixel at all — and the pixels where it is well-conditioned are precisely the corners Harris finds. **One idea with two uses, which is worth noticing because it means 'good feature to track' and 'good feature to match' are the same property**, and Shi and Tomasi's refinement of Harris was derived from exactly that observation.

2 Invariance

Nuisance	Why it breaks naive matching	The standard answer
Scale	A corner photographed at 2 metres is a smooth blur at 20 metres — the patches are not comparable at all.	Detect across a scale pyramid and record the scale at which the response peaked , then describe the patch at that scale. This is the core of SIFT's detector.
Rotation	Patch orientation changes with camera roll, so a raw patch comparison fails.	Estimate a dominant gradient orientation and rotate the patch to a canonical frame before describing it.
Illumination	Brightness and contrast change between exposures and between times of day.	Describe gradients rather than intensities (removing additive brightness) and normalise the descriptor (removing multiplicative contrast).

Nuisance	Why it breaks naive matching	The standard answer
Viewpoint	Surfaces foreshorten: a circular patch becomes an ellipse, and the mapping is not a similarity.	Affine adaptation estimates the local shape, or you accept a limited viewpoint range. Nobody solves this fully , which is why wide-baseline matching stayed hard and is precisely where §4's learned matchers won.
Occlusion	Part of the patch is a different surface entirely, so the descriptor is a blend of two things.	Nothing, at the descriptor level. This is why RANSAC exists — some fraction of matches will simply be wrong and the pipeline must survive it (§3).

A DESCRIPTOR IS A VECTOR DESIGNED TO BE COMPARABLE

Take the patch at the detected scale, rotated to the detected dominant orientation. Scale and rotation have now been removed by construction rather than by the descriptor's own design.

Compute image gradients and histogram their orientations within a 4×4 grid of spatial cells, with 8 orientation bins per cell — 128 numbers. **The spatial histogramming is the most important design decision:** it buys tolerance to small localisation errors, because a gradient that moves by a pixel lands in the same cell.

Normalise the vector, clip any component above 0.2, normalise again. Contrast and brightness are removed by the normalisation, and **the clipping limits the damage a single very strong edge can do** to the whole descriptor.

Compare by Euclidean distance. Every element of the design exists to make that one comparison meaningful — which is worth stating because it is the general principle behind descriptors, including the learned ones of §4, whose training objective is exactly 'make the distance meaningful'.

3 Matching, and surviving the errors

MATCHING gives correspondences that are maybe 40-70% correct. Least squares on that is worthless -- a single gross outlier moves the fit arbitrarily far.

RANSAC:

```
repeat N times:
  sample the MINIMUM set needed to fit the model
    (4 points for a homography, 5 for an essential
     matrix, 3 for a similarity)
  fit the model to exactly those points
  count how many of ALL correspondences agree with it
    within a threshold -- the INLIERS
keep the model with the most inliers
refit on all of that model's inliers
```

HOW MANY ITERATIONS? For inlier fraction w , sample size s , and desired success probability p :

$$N = \log(1 - p) / \log(1 - w^s)$$

$w=0.5, s=4, p=0.99 \rightarrow N = 72$

$w=0.5, s=5, p=0.99 \rightarrow N = 145$

$w=0.5, s=8, p=0.99 \rightarrow N = 1177$

$w=0.3, s=8, p=0.99 \rightarrow N = 70188$

THE SAMPLE SIZE IS IN THE EXPONENT, and so is the inlier fraction. That is why minimal solvers matter: the 5-point essential algorithm is not elegance for its own sake, it is an order-of-magnitude reduction in iterations over an 8-point one -- and why the ratio test, which raises w , pays off exponentially.

THE THREE PARAMETERS THAT DECIDE WHETHER RANSAC WORKS

The inlier threshold. Too tight and true inliers are rejected, so no model gathers support and RANSAC reports nothing; too loose and outliers are admitted, so the final refit is wrong while the inlier count looks healthy. **Set it from your measured feature localisation error** — typically 1 to 3 pixels, and measurable by matching an image against itself under a known transform.

The iteration count. Use the formula, and prefer *adaptive RANSAC*: recompute N from the best inlier fraction observed so far, which terminates far earlier than a fixed conservative N when the data is good.

The ratio test, applied before RANSAC. Keep a match only if the nearest descriptor distance is well below the second nearest — **a ratio under about 0.7 to 0.8. This single filter raises the inlier fraction w substantially, and since w appears in the exponent, it cuts N by orders of magnitude.** It is the cheapest available improvement in the whole pipeline.

And check the final inlier count rather than merely that it returned a model. RANSAC always returns something — it reports the best-supported sample, however poorly supported — and **a model supported by 12 of 800 matches is noise presented through a confident interface**, which is exactly the kind of failure that propagates silently into Module 05's reconstruction.

4 Learned matching

- **Learned descriptors beat SIFT, but only modestly.** This is a genuinely interesting result: **the hand-designed descriptor turned out to be close to optimal for its patch size and task**, and twenty years of design were not easily improved by gradient descent.
- **Learned matchers beat the pipeline decisively.** SuperGlue and its successors take the full set of features from both images and match them *jointly*, using attention (CSCE 636 Module 06) over both sets plus an optimal-transport assignment layer.
- **Because matching is a global assignment problem and nearest-neighbour-with-ratio-test treats it as many independent local ones. The joint formulation is the real contribution**, not the learning as such — a point that becomes clear when you notice the assignment layer is a classical algorithm (CSCE 669 Module 11's matching).
- **The gains are largest at wide baselines and on low-texture surfaces** — precisely the cases §2's table identified as unsolved. **Learning attacked the exact residual weakness**, which is the pattern worth looking for when deciding where learning will help.
- **And the geometry is entirely unchanged. RANSAC, the essential matrix, triangulation, and bundle adjustment all still run on the matcher's output**, exactly as before. **Learning replaced one stage of the pipeline, not the pipeline** — which is the realistic picture of learned geometric vision and a useful corrective to the impression that deep learning displaced it.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Szeliski — Computer Vision, chapter 7 (free PDF)	Feature detection, description, and matching — the reference for §1 through §3.
Lowe — Distinctive Image Features from Scale-Invariant Keypoints (free)	SIFT, in the author's own words. The ratio test of §3 is introduced here, with the distance distributions that justify it.
Fischler & Bolles — Random Sample Consensus (free)	The original RANSAC paper , including the iteration-count derivation of §3.
Sarlin et al. — SuperGlue (free)	The §4 result. The attention-plus-assignment architecture, and the wide-baseline evaluation.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Implement Harris from the structure tensor** and visualise the two eigenvalues as a colour image.
- 2 **Confirm that edges have one large eigenvalue** and corners two, by inspecting known regions.
- 3 **Detect features across a scale pyramid** and verify the same world point is found at the right scale from two distances.
- 4 **Implement a SIFT-like descriptor** and match two images of the same scene.
- 5 **Measure the inlier fraction with and without the ratio test**, at ratios 0.6, 0.7, 0.8, and 0.9.
- 6 **Compute the required RANSAC iterations for each** and report the ratio.
- 7 **Implement RANSAC for a homography** and plot inlier count against threshold.
- 8 **Run RANSAC on deliberately bad matches** (random pairs) and report what it returns.
- 9 **Match a wide-baseline pair with SIFT and with a learned matcher** and compare inlier counts.
- 10 **Find a pair where SIFT fails completely** and explain which invariance broke.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What makes a feature localisable, and what is the aperture problem?
- 2 Derive the Harris response from the structure tensor.
- 3 Where else does the structure tensor appear in this course?

- 4 Name five nuisances and the standard answer to each.
- 5 Which invariance is unsolved, and what follows from that?
- 6 Describe SIFT's descriptor and say what each design step buys.
- 7 State RANSAC's iteration formula and explain why minimal solvers matter.
- 8 Name RANSAC's three parameters and how to set each.
- 9 What did learned matching change, and what did it leave alone?